

MEAN MACHINE
A Rafale Marine on a sortie

At last, after nearly a year of deliberations, the matter of selecting fighter aircraft for India's newly inducted indigenous aircraft carrier, INS *Vikrant*, has been settled. On July 13, when Prime Minister Narendra Modi was on his way to Paris for a two-day visit, a meeting of the Defence Acquisition Council (DAC) convened by defence minister Rajnath Singh granted the Acceptance of Necessity (AoN) to the Indian Navy to acquire 26 Rafale Marine (Rafale-M) fighter jets from France. The deal to procure the Rafale-Ms will cost India at least $8 billion (Rs 65,920 crore) and will be executed through an intergovernmental agreement between India and France. India will acquire 22 single-seater Rafale-Ms and four of its twin-seater trainer versions. They are expected to be based at Visakhapatnam, the home base of INS *Vikrant*. According to military aviators, the Rafale-M is considered among the world's most advanced carrier-borne fighter jets, and its induction in the Indian Navy will add immense capability in terms of its strike, air defence and reconnaissance missions.

The AoN being the first stage of a long-drawn process of defence procurement, officials involved in negotiations claim that it will take months, or even a year, to finalise the Rafale-M deal. Once the negotiations on cost and techno-commercial details are concluded, the deal will require final clearance from the Cabinet Committee on Security chaired by PM Modi before being finally inked. The first Rafale-M is expected to arrive in India at least two-three years after the contract is signed. However, some MoD officials say that given the urgency of the matter, the Indian Navy might ask Rafale manufacturer Dassault Aviation to spare at least four or five Rafale-Ms from its ongoing orders for the French navy.

Though expectations were high that Modi may announce the deal after bilateral discussions with French president Emanuel Macron in Paris, the Rafale-M did not figure in the joint statement by the leaders. Some in the Indian government believe that given the controversy and Opposition criticism following Modi's announcement in 2015 about the acquisition of 36 Rafales for the Indian Air Force, the PM has taken a cautious approach this time, especially considering the fast approaching Lok Sabha election in 2024. However, Dassault Aviation in a press release on July 14 announced that the India had selected the naval variant of Rafale fighter jets for its navy.

The Rafale-M was in contention along with the US's Boeing F/A-18 Super Hornet for the flight deck of INS *Vikrant*. Last year, both fighters displayed their capabilities during trials at the shore-based INS *Hansa* test facility in Goa, which simulated the deck of INS *Vikramaditya*, the navy's other aircraft carrier. The two fighters went through extensive capability trials, including take-off on a ski-jump ramp. This technique, called short take-off but arrested recovery or STOBAR, is used on INS *Vikramaditya* and INS *Vikrant*. In the end, the Rafale-M was deemed more suitable by the navy.

ROAD TO RAFALE

The Indian Navy has been operating the Russian-made Mikoyan-Gurevich MiG-29K fighters from its aircraft carriers ever since the British-made Sea Harriers were phased out after serving for almost three decades. In 2004 and in 2010, 45 MiG-29Ks and associated equipment were ordered at a cost of $2.2 billion (nearly Rs 14,000 crore) in two tranches from Russia. But the MiG-29Ks that operate from INS *Vikramaditya* had a terrible record of accidents, with five aircraft crashing in the past four years, most recently in October 2022. In addition, a 2016 report by the Comptroller and Auditor General (CAG) flagged the occurrence of mid-air engine failures and defects in the airframe of the MiG-29K. "The MiG-29K, which is a carrier-borne multi-role aircraft...